

Suchitra Rajkumar

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SUMMARY

Master of Computer Science Candidate at the Illinois Institute of Technology with professional experience at Big 4. Full-stack developer proficient in building scalable web applications and optimizing backend performance using React, Angular, Node.js, and Spring Boot. Focused on delivering high-quality UI functionalities and efficient data-driven solutions for large-scale clients.

EDUCATION

Master of Computer Science, Computer Science, Illinois Institute of Technology, USA

Aug 2025 – May 2027 (Anticipated)

- GPA:** 3.34
- Relevant Courses:** Advanced Database Organization, Operating Systems, Science of Programming
- Activities:** Executive Board Member for IEEE student organization, Incoming Senior Writer & Copy Editor at *TechNews*

Bachelor of Technology in Computer Science and Engineering,

B S Abdur Rahman Crescent Institute of Science and Technology, IND

Sep 2021 – May 2025

- GPA:** 3.68
- Relevant Courses:** Full Stack Application Development (Frontend & Backend), Big Data and Cloud Computing, Data Structures and Algorithms, AI Techniques, Database Management Systems
- Activities:** President of Crescent Literary Society (2024-25), Editor-in-Charge for *CrescentLine* student newsletter (2023-24)

SKILLS

- Frontend:** React.js, Angular.js, HTML, CSS, JavaScript, TypeScript
- Backend:** Node.js, Spring Boot, Django, Express, PHP
- Languages:** Python, Java, C/C++
- Databases:** MySQL, PostgreSQL, MongoDB, NoSQL
- Tools & Platforms:** AWS, GitHub, CI/CD Pipelines, Agile, MS Suite (Word, Excel, Outlook, PowerPoint)

EXPERIENCE

Frontend Developer Intern – Cyber Risk and Regulatory, PwC, Bangalore, IND

Feb 2025 – Jul 2025

- Contributed to feature development and UI functionalities for a Fortune 100 client (Capital One) under the Financial Services Risk and Regulatory (FS R&R) team
- Increased regulatory data visibility by 40% as measured by compliance team feedback, by developing responsive **Angular-based dashboards** for cross-functional compliance teams
- Enhanced dashboard interactivity and usability by building reusable **Angular components** with real-time data filtering, sorting, and export capabilities for compliance reporting workflows
- Improved user interface load times as evaluated by performance benchmarks, by implementing lazy loading, component optimization, and efficient state management in the **Angular frontend architecture**
- Delivered 5+ frontend features as quantified by sprint completions, by collaborating with backend engineers and UX designers to translate regulatory requirements into functional, user-friendly interfaces

Full Stack Development Intern, vTitan Corporation Pvt Ltd, Chennai, IND

Jul 2024

- Improved data management efficiency by reducing average task completion time by 35%, by designing and delivering a responsive **React** UI for internal systems using **JavaScript, HTML, and CSS**
- Decreased API response time by 30% as measured by load testing benchmarks by building and deploying **RESTful APIs** with **Spring Boot** and **MySQL** for real-time data processing
- Increased database query throughput as computed by performance metrics, by debugging and optimizing **MySQL** relational models through indexing and query tuning

Software Developer Intern, Bluestock Fintech, Remote, IND

Jul 2024 – Aug 2024

- Delivered 100% functional fintech security dashboard as assessed by acceptance criteria, by implementing an end-to-end Python Django application with real-time **data visualization** and CRUD operations
- Improved backend query performance by debugging and refactoring **Django** ORM queries and optimizing database workflows
- Enhanced user experience through responsive **JavaScript, HTML, and CSS** frontend design supporting secure multi-device accessibility

PROJECTS

RuralReach: Deep Learning System for Hospital Referrals (Academic Project)

Oct 2024 – Apr 2025

- Achieved 85% prediction accuracy for optimized hospital routing by developing AI-driven neural models using **Python** and **TensorFlow**
- Built a full-stack solution using **React** and **Node.js**, enabling real-time data visualization with sub-second response times
- Analyzed the impact of optimized routing on simulated patient outcomes, projecting a significant reduction in readmission rates based on model predictions

B+-Tree Index Manager Implementation

Dec 2025

- Implemented database-grade B+ tree index in C, achieving O(log n) insert, delete, and search operations with disk-backed persistence and automatic tree rebalancing
- Built a buffer manager supporting FIFO, LRU, and CLOCK page replacement strategies with dirty page tracking and force-flush capabilities
- Designed a recursive node splitting algorithm propagating splits upward through parent nodes, dynamically growing tree height as needed
- Passed a comprehensive test suite validating insertion, deletion, search, and range scan correctness under various workload patterns